

chapter 6

Linear Momentum And Energy

9. Perfectly Inelastic Collision in One Dimension

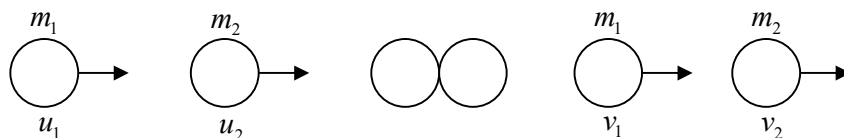
Final Velocity

When perfectly inelastic bodies moving along the same line collide, they stick to each other. Let m_1 and m_2 be the masses, v_1 and v_2 be their velocities before the collision and v be the common velocity of the bodies after the collision. By the conservation of linear momentum,

$$m_1 v_1 + m_2 v_2 = m_1 v + m_2 v$$

or
$$v = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2}$$

Coefficient of Restitution (e)



When 2 masses m_1 and m_2 collide there is deformation of the mass. As collision is not elastic and the masses try to regain their shapes, some energy is lost as bodies are not perfectly elastic. Therefore reducing their velocities. For perfectly elastic bodies, absence of friction leads to conservation of energy and bodies regain their shapes.

This coefficient is defined as

$$e = \left| \frac{\text{relative velocity of separation}}{\text{relative velocity of approach}} \right| \Rightarrow e = \left| \frac{v_2 - v_1}{v_1 - v_2} \right|$$

e is usually a positive, real number between 0 and 1:

$e = 0$: This is a perfectly *inelastic* collision. This means kinetic energy along the common normal is 0. Kinetic energy is converted to heat or work done in deforming the objects.

$0 < e < 1$: This is a real-world *inelastic* collision, in which some kinetic energy is dissipated.

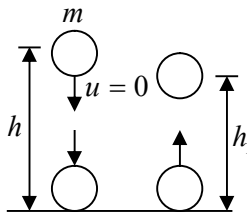
$e = 1$: This is a perfectly *elastic* collision, in which no kinetic energy is dissipated, and the objects rebound from one another with the same relative speed with which they approached.

Example: A ball of mass m strikes a surface falling freely from height h . The coefficient of restitution between ball and surface is e .

(a) Find the total distance before it stops.

(b) Find the total time elapsed before it stops.

Solution: (a) Velocity of ball just before striking surface



A ball of mass m strikes a surface falling freely from height h . The coefficient of restitution between ball and surface is e . Velocity of ball just before striking surface

$$v = \sqrt{2gh}$$

The velocity with which ball rises

$$v_1 = e\sqrt{2gh} = ev$$

It goes to a height h_1

$$\text{as } 0 = (ev)^2 - 2gh_1 \Rightarrow h_1 = \frac{e^2 \times 2gh}{2g} = e^2 h$$

After second rebound, velocity $v_2 = ev_1 = e^2 v$ and it goes to a height $h_2 = e^2 h_1 = e^4 h$

After n^{th} rebound $h_n = e^{2n} h$ and $v_n = e^n v$

Total distance moved by ball till it stops

$$\begin{aligned} H &= h + 2h_1 + 2h_2 + \dots \infty \text{ terms} = h + 2(e^2 h + e^4 h + \dots \infty \text{ terms}) \\ &= h + 2e^2 (1 + e^2 + \dots \infty \text{ terms}) \\ &= h + 2e^2 h \frac{1}{1 - e^2} \Rightarrow H = h \times \left(\frac{1 + e^2}{1 - e^2} \right) \quad \left(s = \frac{a}{1 - r}, |r|^2 \text{ for an infinite G.P.} \right) \end{aligned}$$

(b) Time taken by the ball to reach the ground from height h , $t = \sqrt{\frac{2h}{g}}$

$$\text{For 1st rebound, } t_1 = \sqrt{\frac{2h_1}{g}} = \sqrt{\frac{2 \times e^2 h}{g}} \Rightarrow t_1 = et = e \sqrt{\frac{2h}{g}}$$

$$\text{Similarly, for 2nd rebound } t_2 = e + 1 = e^2 \sqrt{\frac{2h}{g}}$$

Total time taken by ball before stopping

$$\begin{aligned} T &= t + (2t_1 + 2t_2 + \dots \infty \text{ terms}) \\ &= \sqrt{\frac{2h}{g}} + 2e \sqrt{\frac{2h}{g}} (1 + e + e^2 + \dots \infty) = \sqrt{\frac{2h}{g}} \left[1 + 2e \frac{1}{1 - e} \right] = \sqrt{\frac{2h}{g}} \left(\frac{1 + e}{1 - e} \right) \end{aligned}$$

For $e = 1$, collision is perfectly plastic,

$$T = \sqrt{\frac{2h}{g}} \quad (\text{ball does not bounce after rebound})$$

for $e = 1$, $T \rightarrow \infty$, ball keeps on bouncing to same height indefinitely as no energy is lost.